

Parth Aggarwal

M.Tech. Structural and Computational Biology

Department of Biosciences and Bioengineering, Indian Institute of Technology Roorkee

parth_a@bt.iitr.ac.in · github.com/agpparth

Research Interests

Cryo-electron microscopy (cryo-EM) and cryo-electron tomography (cryo-ET); deep learning for structural biology; implicit neural representations for heterogeneous reconstruction, alignment and pose estimation.

Education

M.Tech. in Structural and Computational Biology

Indian Institute of Technology Roorkee, Roorkee

2025 – 2027 (expected May 2027)

CGPA 9.231 / 10 (39 credits completed)

B.Tech. in Biotechnology

Delhi Technological University (DTU), New Delhi

2021 – 2025

CGPA 8.10 / 10, 174 credits

Medium of instruction throughout: English.

Research Experience

Graduate Researcher — Singla Lab

June 2026 – present

Dept. of Biosciences and Bioengineering, IIT Roorkee Supervisor: Prof. Jitin Singla

- Surveying deep learning methods for cryo-EM and cryo-ET: implicit neural representation architectures and activation functions, including SECO-INR, and the CryoDRGN line of approaches to heterogeneous reconstruction. First author on a comprehensive survey arising from this work (in preparation; preprint expected October 2026).
- Current project: Developing a reference-free subtomogram alignment method for cryo-electron tomography that jointly aligns particles by gradient ascent on a missing-wedge-aware spectral SNR score (SFSC).

B.Tech. Thesis — Review of AI and Nanotechnology used for Microbiome engineering

Aug 2024 – May 2025

Dept. of Biotechnology, DTU Supervisor: Dr. Prakash Chandra

- Surveyed applications of machine learning and nanotechnology to microbiome engineering, covering predictive modelling of microbial interactions and nanodevice-based intervention strategies.

Publications

- Aggarwal, P., Singla, J. “Deep learning methods in cryo-EM and cryo-ET: a survey.” *In preparation*, 2026.

Selected Projects

cryoET-Heterogeneity — validation layer for CryoDRGN-ET latent spaces

Sep 2026

Independent project github.com/agpparth/cryoET-Heterogeneity

- Python package adding a quantitative analysis layer over CryoDRGN-ET latent embeddings: dimensionality reduction, clustering swept over K with three quality metrics, representative-state extraction as real observed latent points rather than arithmetic centroids, k -nearest-neighbour trajectory analysis, and null-model validation.
- On a real 4,549-particle, 8-dimensional embedding: the latent space exceeds a covariance-matched Gaussian null by 26.5–77.1 null standard deviations at every K , but no particular K is supported — four criteria give four answers. A structureless correlated Gaussian returns a *higher* silhouette (0.53) than anything measured on the real data (0.27–0.29), which is why a bare silhouette cannot support a claim.

- 51 unit tests; bit-reproducible across independent runs after eliminating two sources of nondeterminism that a fixed seed does not control (multithreaded BLAS reductions in covariance/SVD, and OpenMP work-splitting in K -means).

cfDNA V-Plot Generator

Nov 2025

IIT Roorkee github.com/agpparth

- Python and R pipeline that bins cell-free DNA fragments by distance from motif centre and by fragment length from TSV alignment records, and renders a V-plot heatmap (fragments ≤ 200 bp) in ggplot2.

TCGA RNA-seq Differential Expression Analysis

Nov 2025

IIT Roorkee github.com/agpparth

- Bash and R pipeline extracting NKX2-1 expression (TPM) across a TCGA cohort from GDC manifest files, comparing primary tumour against solid tissue normal as $\log_2(\text{TPM} + 1)$ boxplots.

Technical Skills

Programming	Python, R, C, JavaScript, Bash
Machine learning	PyTorch, scikit-learn, NumPy, SciPy, UMAP
Data and plotting	matplotlib, ggplot2, tidyr, pandas
Structural biology	CryoDRGN / CryoDRGN-ET, Schrödinger, CCP4
Tools	Linux, Git, pytest
Languages	English (full professional), Hindi (native), Punjabi

Awards and Scholarships

- **GATE 2025, Biotechnology** — All India Rank 124 of 20,048 candidates (top 0.62%), score 695/1000.
- **GATE 2024, Biotechnology** — All India Rank 327 of 17,078 candidates (top 1.91%), score 590/1000.
- **MHRD Scholarship**, Government of India — awarded on GATE qualification for M.Tech. study at IIT Roorkee (2025–2027).

Relevant Coursework

M.Tech., IIT Roorkee: Machine Learning Essentials (A); Computational Biology (A+); Protein Design and Engineering (A+); Structural Biology (A); Bioanalytical Techniques (A+); Biomolecular Interactions and Drug Design (A); Computer Programming (A).

Additional Experience

Student Mentor Volunteer

Feb – Apr 2022

Desh Ke Mentor, Government of NCT of Delhi

- Mentored school students on academic planning and career decisions.

References

Prof. Jitin Singla

Assistant Professor

Dept. of Biosciences and Bioengineering

Indian Institute of Technology Roorkee

Roorkee, Uttarakhand 247667, India

jsingla@bt.iitr.ac.in

jitinsingla.in

M.Tech. research supervisor